

Bayesian Learning

Lecture 8 - Markov Chain Monte Carlo. Metropolis-Hastings.

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Lecture overview

- Markov Chain Monte Carlo
- Metropolis-Hastings
- MCMC - efficiency, burn-in and convergence

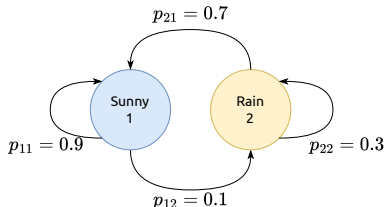
Stochastic process

- Let $\mathcal{S} = \{1, 2, \dots, K\}$ be a finite set of **states**.
 - ▶ Weather: $\mathcal{S} = \{\text{sunny}, \text{rain}\}$.
 - ▶ School grades: $\mathcal{S} = \{A, B, C, D, E, F\}$
- **Stochastic process**: collection of random variables $X_1, X_2, X_3, \dots$, often over time.
- A **time series** with categories.
- Weather: $X_1 = \text{sunny}, X_2 = \text{sunny}, X_3 = \text{rain}, X_4 = \text{sunny}$.
- School grades: $X_1 = C, X_2 = C, X_3 = B, X_4 = A, X_5 = B$.

Markov chains

- **Markov chain:** probability distribution of tomorrow's state X_{t+1} depends (only) on today's state X_t :

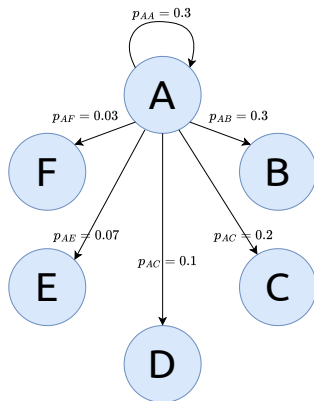
$$p_{ij} = \Pr(X_{t+1} = j | X_t = i)$$



- **Transition matrix** for weather example

$$\mathbf{P} = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} = \begin{pmatrix} 0.9 & 0.1 \\ 0.7 & 0.3 \end{pmatrix}$$

Markov chains - Grades example



$$P = \begin{matrix} & \begin{matrix} A & B & C & D & E & F \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \\ E \\ F \end{matrix} & \begin{pmatrix} 0.3 & 0.3 & 0.2 & 0.1 & 0.07 & 0.03 \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ . & . & . & . & . & . \end{pmatrix} \end{matrix}$$

Stationary distribution

■ h -step transition probabilities

$$p_{ij}^{(h)} = \Pr(X_{t+h} = j | X_t = i)$$

■ h -step transition matrix by matrix power

$$\mathbf{P}^{(h)} = \mathbf{P}^h$$

■ Unique equilibrium distribution $\pi = (\pi_1, \dots, \pi_k)$ if chain is ergodic:

- ▶ **irreducible** (possible to get to any state from any state)
- ▶ **aperiodic** (does not get stuck in predictable cycles)
- ▶ **positive recurrent** (expected time of returning is finite)

■ Limiting long-run distribution as $h \rightarrow \infty$

$$\mathbf{P}^h \rightarrow \begin{pmatrix} \pi \\ \pi \\ \vdots \\ \pi \end{pmatrix} = \begin{pmatrix} \pi_1 & \pi_2 & \cdots & \pi_k \\ \pi_1 & \pi_2 & \cdots & \pi_k \\ \vdots & \vdots & & \vdots \\ \pi_1 & \pi_2 & \cdots & \pi_k \end{pmatrix}$$

Stationary distribution, cont.

- Limiting **long-run distribution** as $h \rightarrow \infty$

$$\mathbf{P}^h \rightarrow \begin{pmatrix} \pi \\ \pi \\ \vdots \\ \pi \end{pmatrix} = \begin{pmatrix} \pi_1 & \pi_2 & \cdots & \pi_k \\ \pi_1 & \pi_2 & \cdots & \pi_k \\ \vdots & \vdots & & \vdots \\ \pi_1 & \pi_2 & \cdots & \pi_k \end{pmatrix}$$

- Stationary distribution**

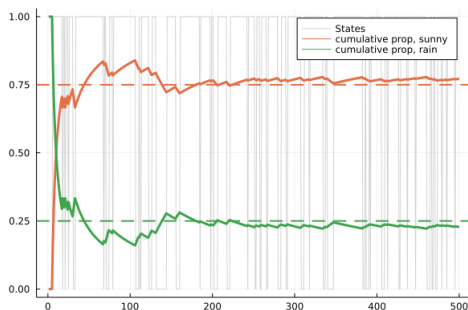
$$\pi = \pi \mathbf{P}$$

- Draw starting point using π . All future states of the Markov Chain will have distribution π .
- Weather example:

$$\mathbf{P} = \begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix}, \mathbf{P}^2 = \begin{pmatrix} 0.84 & 0.16 \\ 0.42 & 0.58 \end{pmatrix}$$
$$\mathbf{P}^5 = \begin{pmatrix} 0.77 & 0.23 \\ 0.69 & 0.31 \end{pmatrix}, \mathbf{P}^{100} = \begin{pmatrix} 0.75 & 0.25 \\ 0.75 & 0.25 \end{pmatrix}$$
$$\pi = (0.75, 0.25)$$

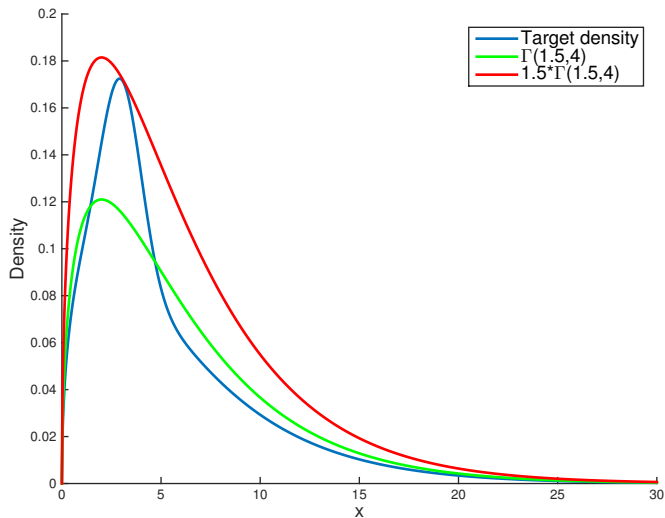
The basic MCMC idea

- Aim: simulate from a discrete distribution $p(x)$.
- **MCMC**: simulate a **Markov Chain** with a **stationary distribution** that is exactly $p(x)$.
- How to set up the transition matrix P ? **Metropolis-Hastings!**



- Can be extended to **continuous random variables**.

Rejection sampling



Random walk Metropolis algorithm

■ **Initialize** $\theta^{(0)}$ and iterate for $i = 1, 2, \dots$

1 **Sample proposal:** $\theta^* | \theta^{(i-1)} \sim N(\theta^{(i-1)}, c \cdot \Sigma)$

2 Compute the **acceptance probability**

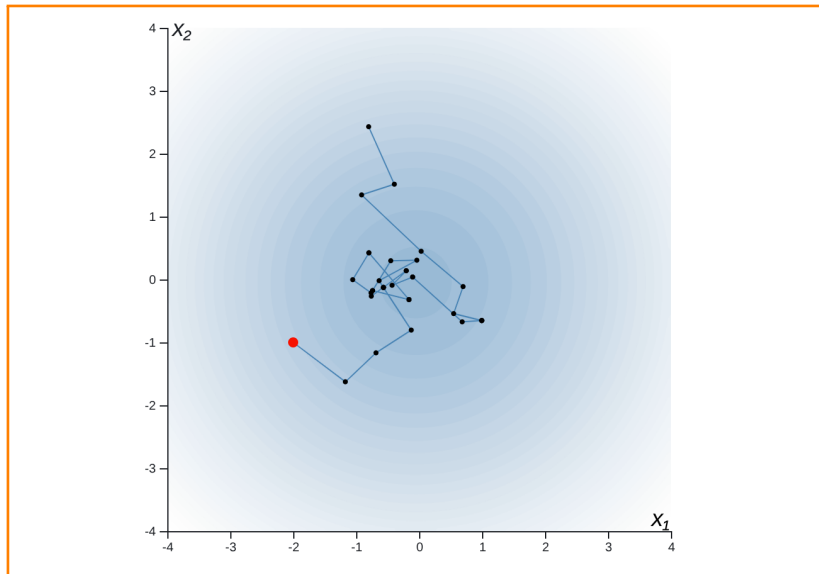
$$\alpha = \min \left(1, \frac{p(\theta^* | \mathbf{y})}{p(\theta^{(i-1)} | \mathbf{y})} \right)$$

3 Draw $u \sim \text{Uniform}(0, 1)$

If $u \leq \alpha$ set $\theta^{(i)} = \theta^*$ (accept and move)

If $u > \alpha$ set $\theta^{(i)} = \theta^{(i-1)}$ (reject and stay)

Interactive - Random Walk Metropolis



Proportional form of posterior is enough!

- Assumption: we can compute $p(\boldsymbol{\theta}|\mathbf{y})$ for any $\boldsymbol{\theta}$.
- Proportionality constants in posterior cancel out in

$$\alpha = \min \left(1, \frac{p(\boldsymbol{\theta}^*|\mathbf{y})}{p(\boldsymbol{\theta}^{(i-1)}|\mathbf{y})} \right).$$

- **Proportional form of posterior is enough!**

$$\alpha = \min \left(1, \frac{p(\mathbf{y}|\boldsymbol{\theta}^*) p(\boldsymbol{\theta}^*)}{p(\mathbf{y}|\boldsymbol{\theta}^{(i-1)}) p(\boldsymbol{\theta}^{(i-1)})} \right)$$

Random walk Metropolis - choosing a proposal

- Common choices of Σ in proposal $N(\theta^{(i-1)}, c \cdot \Sigma)$:
 - ▶ $\Sigma = I$ (proposes 'off the cigar')
 - ▶ $\Sigma = J_y^{-1}(\tilde{\theta})$ (propose 'along the cigar')
 - ▶ **Adaptive**. Start with $\Sigma = I$. Update Σ from initial run.
- Set c so average acceptance probability is 25-30%.
- **Good proposal**:
 - ▶ **Easy to sample**
 - ▶ **Easy to compute** α
 - ▶ Proposals should take reasonably **large steps** in θ -space
 - ▶ Proposals should **not be reject too often**.

The Metropolis-Hastings algorithm

- Generalization when the proposal density is not symmetric.

- Initialize $\theta^{(0)}$ and iterate for $i = 1, 2, \dots$

1 **Sample proposal:** $\theta^* \sim q(\cdot | \theta^{(i-1)})$

2 Compute the **acceptance probability**

$$\alpha = \min \left(1, \frac{p(\theta^* | \mathbf{y})}{p(\theta^{(i-1)} | \mathbf{y})} \frac{q(\theta^{(i-1)} | \theta^*)}{q(\theta^* | \theta^{(i-1)})} \right)$$

3 Draw $u \sim \text{Uniform}(0, 1)$

If $u \leq \alpha$ set $\theta^{(i)} = \theta^*$ (accept and move)

If $u > \alpha$ set $\theta^{(i)} = \theta^{(i-1)}$ (reject and stay)

The independence sampler

- **Independence sampler:** $q(\theta^* | \theta^{(i-1)}) = q(\theta^*)$.

- **Proposal** is **independent of previous draw**.

- Example:

$$\theta^* \sim t_v(\tilde{\theta}, J_y^{-1}(\tilde{\theta})),$$

where $\tilde{\theta}$ and $J_y(\tilde{\theta})$ are computed by numerical optimization.

- Can be very **efficient**, but has a tendency to **get stuck**.

- Make sure that $q(\theta^*)$ has **heavier tails** than $p(\theta|y)$.

The efficiency of MCMC

- **How efficient** is MCMC compared to iid sampling?
- If $\theta^{(1)}, \theta^{(2)}, \dots, \theta^{(N)}$ are **iid** with variance σ^2 , then

$$\text{Var}(\bar{\theta}) = \frac{\sigma^2}{N}.$$

- Autocorrelated $\theta^{(1)}, \theta^{(2)}, \dots, \theta^{(N)}$ generated by **MCMC**

$$\text{Var}(\bar{\theta}) = \frac{\sigma^2}{N} \left(1 + 2 \sum_{k=1}^{\infty} \rho_k \right)$$

where $\rho_k = \text{Corr}(\theta^{(i)}, \theta^{(i+k)})$ is the autocorrelation at lag k .

- **Inefficiency factor**

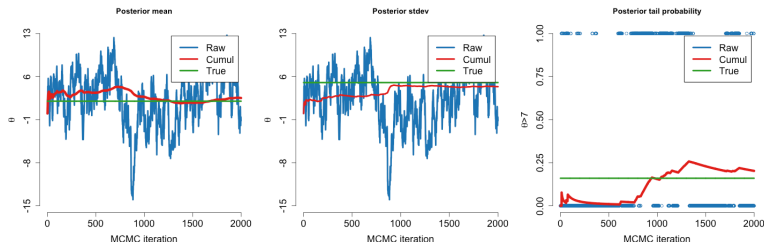
$$\text{IF} = 1 + 2 \sum_{k=1}^{\infty} \rho_k$$

- **Effective sample size** from MCMC

$$\text{ESS} = N/\text{IF}$$

Burn-in and convergence

- How long **burn-in**?
- How long to **sample** after burn-in?
- **Convergence diagnostics**
 - ▶ Raw plots of simulated sequences (**MCMC trajectories**)
 - ▶ **Cumulative estimates** plots
 - ▶ **Repeated runs** with different initial values
 - ▶ Potential scale reduction factor, R .



Burn-in and convergence

